

Parameterization on formation free energy of dislocation loops up to 1100 K in bcc iron

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ARTICLE INFO

Article history:

Received 17 May 2021

Revised 3 November 2021

Accepted 13 November 2021

Available online 18 November 2021

Keywords:

Iron

Dislocation loops

Formation free energy

Anisotropic elasticity

Analytical formula

ABSTRACT

This work provides a simple analytical formula with a set of best-fit coefficients for evaluating formation free energies of prismatic dislocation loops as a function of temperature in body-centered-cubic iron. The parameterization is made based on the anisotropic elastic self-energies and corresponding non-linear core-energies of edge dislocations. The variation of these two energy terms with temperature is estimated by including temperature-dependent elastic stiffness tensors in the anisotropic elasticity calculation, and based on the stability of C15 clusters and dislocation loops, respectively. The new analytical formula explicitly contains temperature and is valid for temperatures up to 1100 K.

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Prismatic dislocation loops are the primary manifestation of irradiation damage in crystals, and contribute to the irradiation hardening and embrittlement [1–5]. Self-interstitial atoms (SIAs) and vacancies produced by irradiation are basic ingredients for the formation of dislocation loops. Usually, the aggregation of single type defects gives rise to loop nucleation. Thereafter, loops grow or shrink by absorbing and/or emitting further defects. The strength of absorption and emission has a dependence on the energetics of evolving dislocation loops. For instance in cluster dynamics simulations, defect emission rate of loops is determined by the loop binding free energy (to the emitted defect), which is closely correlated to the formation free energy of loops [6]. For such simulations, formation free energies of loops are significant input parameters. For other simulation methods like kinetic Monte Carlo simulations and mean-field approximations, loop energetics also have important applications [7–9].

Free energies of SIAs aggregators with different structures in bcc iron can be modeled based on *ab initio* methods, for instance, the typical work done by Alexander [10]. However, Alexander does not consider the entropy contributions [11,12], which prove to be difficult to handle especially for structures causing high levels of local deformation [13]. Specifically, the large distortions in the loop core areas (up to deformation of 15% [14]) limit the quantitative accuracy of magnetic moment prediction of atoms in close proximity to the interstitial defects [13]. For such strongly distorted

regions, there presently exists no calculational method to compute the temperature-dependent free energy due to phonons and magnons [13,15].

Extensive experiments have shown that temperature strongly influences the behavior of dislocation loops, leading to the variation of their size, shape and even type [2,16,17]. For iron, one of distinguishing features related to temperature effects is the occurrence of the [001] type dislocation loops at high temperatures [16,17]. This phenomenon does not occur in most bcc metals because, according to the conventional isotropic treatment of elasticity [18] and molecular dynamics (MD) simulations (Fig. 1 in [19]), the $1/2\langle 111 \rangle$ configuration should prefer to form due to a lower self-energy. The mechanism that [001] loops form via the interaction between two migrating $1/2\langle 111 \rangle$ loops [20,21], seems not able to explain the fact that [001] loops are rarely observed in non-magnetic bcc metals, where small prismatic $1/2\langle 111 \rangle$ loops are just as mobile as in bcc iron [15]. Recently, simulations reported the formation of C15 clusters prior to any loops formation and their collapse into loops of both types [19,22]. After loops formation, their thermodynamic stabilities were reported to be associated with the strong softening of shear stiffness constants C' in bcc iron at elevated temperatures [23]. Further studies showed that C' reduction has a profound effect on the anisotropic elastic free self-energies of dislocations in iron, in turn, on the relative stability of dislocation loops [15]. To assess the free energy of dislocation loops in iron, especially for large loops observable experimentally, the full anisotropic elasticity approximation might be the necessary choice.

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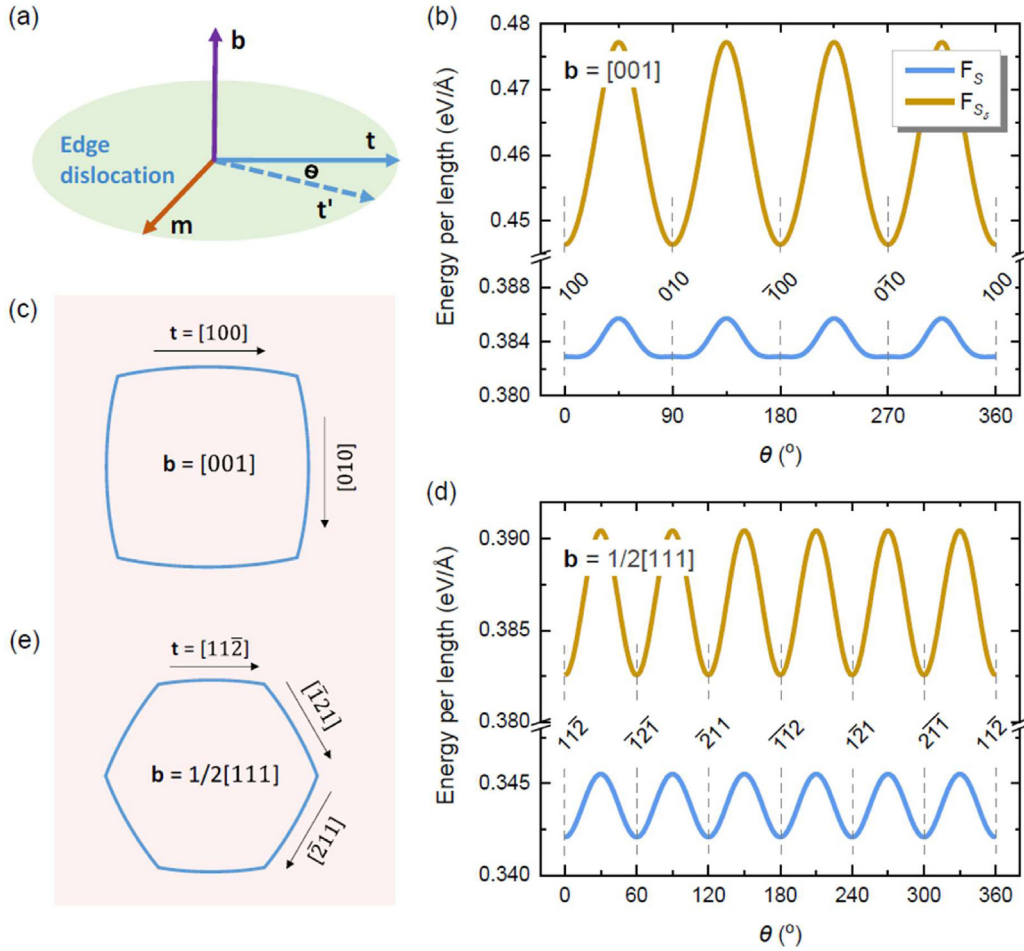


Fig. 1. Elastic energy of straight dislocations and its implication on loop configuration. (a) Schematic drawing shows the rotation of dislocations around the direction of Burgers vector. (b)(d) Variation of the elastic energy F_S and F_{S_s} of straight dislocations, and (c)(e) corresponding favorable configuration of dislocation loops with $\mathbf{b} = \alpha[001]$ and $\alpha/2[111]$ in bcc iron at 0 K.

In this work, we calculate elastic energies of dislocation loops in iron as a function of temperature up to 1100 K based on the anisotropic elasticity of dislocations. Free energies of loops are assessed by adding the elastic energy and core energy of dislocation segments constituting the sides of the loop. For an arbitrary dislocation in a force free body, the total linear elastic energy is

$$E_e = \frac{1}{2} \int_V C_{ijkl} u_{i,j}(\mathbf{x}) u_{k,l}(\mathbf{x}) dV \quad (1)$$

where C_{ijkl} is the temperature-dependent elastic stiffness tensor. $u_{i,j}$ is the elastic distortion field tensor of the dislocation at an arbitrary point $\mathbf{x}$ in the body. V is the volume of the complete body excluding the cylindrical core region of radius δ surrounding the dislocation, given that at closer distances to the dislocation the distortions exceed the range of validity of the linear distortion theory.

For a closed loop and finite body, the volume integral of Eq. (1) can be converted to two terms of surface integral using Gauss's theorem [24]

$$E_e = \frac{1}{2} \int_S C_{ijkl} b_i n_j(\mathbf{x}) u_{k,l}(\mathbf{x}) dS + \frac{1}{2} \int_{S_s} C_{ijkl} u_{i,j}(\mathbf{x}) u_{k,l}(\mathbf{x}) dS_j, \quad (2)$$

where the first integral is over the plane surface S enclosed by the loop but terminating a distance of δ from it. b_i and n_j are the components of the Burgers vector of the loop and its normal vector, respectively. The second integral is over the curved surface S_j surrounding the loop, known as the core-traction energy E_{S_s} . This

part of energy contribution can be evaluated based on the energy of straight finite dislocations constituting the sides of the loop.

Including the contribution of the non-linear core energy E_c , total free energy of a dislocation loop with sides $\mathbf{l}$ can be calculated as [15,24]

$$E_{tot} = \sum_{\mathbf{l}} E_S(\mathbf{l}) \ln\left(\frac{4R}{e\delta}\right) + \sum_{\mathbf{l}} E_{S_s}(\mathbf{l}) + \sum_{\mathbf{l}} E_c(\mathbf{l}) \quad (3)$$

where $E_S(\mathbf{l})$ is the prelogarithmic energy factor of a straight dislocation with length $\mathbf{l}$. R is the radius of the equivalent circular loop, $e \approx 2.7182$ and $\delta \approx 0.4$ nm. For a straight dislocation with the Burgers vector $\mathbf{b}$ and orientation $\mathbf{t} = \mathbf{l}/l$, $E_S(\mathbf{l})$ can be estimated as [15,24]

$$E_S(\mathbf{l}) = l \cdot F_S(\mathbf{t}) = \frac{l}{2\pi} b_i b_m n_j n_q C_{ijkl} C_{npmq} \cdot \text{Im} \left[\sum_{\eta=1}^3 \frac{\kappa_p^{(\eta)} \kappa_l^{(\eta)} N_{kn}(\chi^{(\eta)})}{n_r \frac{\partial D}{\partial \kappa_r}(\chi^{(\eta)})} \right] \quad (4)$$

where $\chi^{(\eta)} = \mathbf{m} + \mathbf{n}\omega^{(\eta)}$, $\mathbf{n} = \mathbf{b}/b$ and $\mathbf{m} = \mathbf{n} \times \mathbf{t}$. $\omega^{(\eta)}$ are the three complex roots (with a positive imaginary part) of the sextic equation $S(\omega) = \det[C_{ijkl}(m_j + n_j\omega)(m_l + n_l\omega)] = 0$. $N_{kn}(\chi)$ is the matrix adjoint to $L_{ik}(\chi) = C_{ijkl} \kappa_j \kappa_l$, which is the inverse of the Fourier transform of the real space anisotropic Green's tensor. $D(\chi) = \det[L_{ik}(\chi)]$. Im denotes "imaginary part of". The core-traction energy E_{S_s} can be evaluated similarly as [24]

$$E_{S_s}(\mathbf{l}) = l \cdot F_{S_s}(\mathbf{t}) = -\frac{l}{2} \int_0^{2\pi} C_{ijkl} (m_j \cos \theta + n_j \sin \theta) H_{ilk}(\mathbf{l}, \theta) d\theta \quad (5)$$

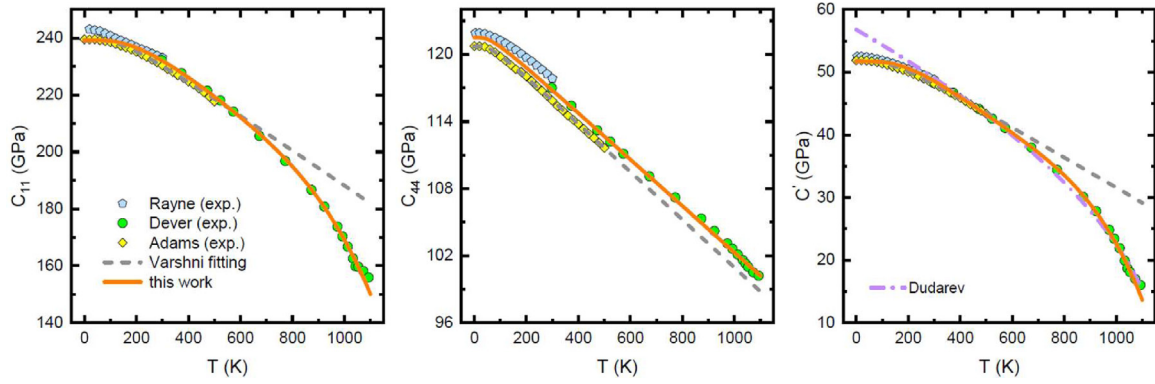


Fig. 2. Three independent elastic constants C_{11} , C_{44} and C' of iron as a function of temperature. The experimental data by Rayne [25], Dever [26] and Adams [27] cover the temperature range of 3 – 1100 K. The dashed line represents a fit of the Varshni expression Eq. (9) to experimental data of Adams. The solid line displays the results of the Varshni fit using the optimized parameters in Table 1. A fit with $C' = 56.8(1 - T/1185)^{1/2}$ GPa by Dudarev [15] is also shown for comparison.

with

$$H_{ilk}(\mathbf{l}, \theta) = \text{Im} \left[\sum_{\eta=1}^3 F_i(\chi^{(\eta)}) \ln(\cos \theta + \omega^{(\eta)} \sin \theta) \right] \times \text{Im} \left[\sum_{\zeta=1}^3 \frac{\kappa_i^{(\zeta)} F_k(\chi^{(\zeta)})}{\cos \theta + \omega^{(\zeta)} \sin \theta} \right] \quad (6)$$

$$F_i(\chi^{(\eta)}) = \frac{1}{\pi} b_f C_{lmjk} n_k \frac{\kappa_m^{(\eta)} N_{li}(\chi^{(\eta)})}{n_r \frac{\partial D}{\partial \kappa_r}(\chi^{(\eta)})} \quad (7)$$

To estimate the total free energy of a loop, a precursor is to know the loop configuration, i.e., configuration of loop sides in the anisotropic iron. For a prismatic dislocation loop with Burgers vector $\mathbf{b}$, the dislocation segments constituting the loop sides are situated in the plane normal to $\mathbf{b}$ with all possible orientations $\mathbf{t}$ around it (Fig. 1(a)). Fig. 1(b) shows the prelogarithmic energy factor and core-traction energy per unit length, $F_5(\mathbf{t})$ and $F_8(\mathbf{t})$ in Eqs. (4) and (5), evaluated for $\mathbf{b} = \alpha[001]$ edge dislocations as a function of rotation angle at 0 K, where α is the lattice parameter in bcc iron. The rotation starts from the initial orientation $\mathbf{t} = [100]$. Clearly, iron displays strong anisotropy already at 0 K. The dislocations with lowest elastic energy are ones with orientation along $[100]$, $[010]$, $[\bar{1}00]$ and $[0\bar{1}0]$ directions. This anisotropy phenomenon potentially indicates a square shape when a $\alpha[001]$ dislocation loop forms, as shown in Fig. 1(c). The square $\alpha[001]$ loops with sides parallel to $\langle 100 \rangle$ directions have been confirmed experimentally [17]. In the case of dislocations with $\mathbf{b} = \alpha/2[111]$, Figs. 1(d) and (e) indicate that dislocation loops of a hexagonal shape with sides parallel to $\langle 11\bar{2} \rangle$ directions would be favorable. However, no experiments has confirmed this predication. Note that $\alpha/2[111]$ loops with sides parallel to $\langle 1\bar{1}0 \rangle$ directions would also be hexagonal. In any case, the free energy of a sided loop for a given $\mathbf{b}$ can be determined by a set of dislocations with equivalent orientations by symmetry [15,24]

$$E_{tot} = P F_5(\mathbf{t}) \ln \left(\frac{4R}{e\delta} \right) + P F_8(\mathbf{t}) + P F_c(\mathbf{t}) \quad (8)$$

where P is the perimeter of the loop, and $F_c(\mathbf{t})$ is the non-linear core energy per unit length.

The temperature dependence of elastic constants in the initial formula Eq. (1) makes it possible to estimate the loop free energy as a function of temperature. For bcc iron, the independent elastic constants are C_{11} , C_{44} and $C' = (C_{11} - C_{12})/2$. Rayne [25], Dever [26] and Adams [27] reported experimental values of elastic constants in bcc iron, covering temperatures from 3 K to 1100 K. Fig. 2 shows that the three sets of experimental data obviously deviate

from each other, especially for C_{11} and C_{44} . Using the semiempirical Varshni expression, Adams fitted the experimental data in the range of 3 – 500 K with [27],

$$C(T) = C_0 - \frac{s}{\exp \left(\frac{T_E}{T} \right) - 1} \quad (9)$$

where C_0 denotes the value of the elastic constant at zero temperature, T_E the effective Einstein temperature, s an adjustable parameter related to the strength of the anharmonic interactions. Though the Varshni fit agrees well with the data of Adams up to 500 K [27], it clearly deviates from other sets of data above 500 K (Fig. 2). To extend the valid range, we assume that the s parameter in Eq. (9) is also temperature-dependent, to account for possible temperature effects on anharmonic interactions. The new fit turns out to be rather good. It predicts successfully the experimental data in a wider range towards 1100 K (Fig. 2). The parameters yielded from the fit are listed in Table 1.

With above formulas and elastic constants, the prelogarithmic energy factor and core-traction energy of dislocations can be conveniently evaluated as a function of temperature according to Eq. (8). Note that the calculations of the two terms do not explicitly include entropic contributions. However, elastic constants are defined in terms of second-order variations with respect to a system free energy and therefore the use of experimental elastic constant values as a function of temperature implicitly includes lattice entropic effects due to phonons and magnons [15]. The determination of total free energy of a loop requires the evaluation of core energy E_c as a function of temperature. In principle, the core energy of dislocation can be determined by comparing the elastic energy of dislocation with the total energy derived from atomistic simulations, but only for the case at zero temperature. Unfortunately, no computational methods are presently available to quantitatively account for temperature effects on the core energy term. For this, Dudarev suggested an approximation that the core energy of a dislocation scales similarly with respect to temperature as that of the core-traction energy [15]. It implies that the zero-temperature core energy acts as the reference energy for the corresponding core energy terms at higher temperatures. Precise determination of these values will be important. The difficulty is that zero-temperature core energies deduced from atomic simulations strongly depend on the choice of interatomic potential. For dislocation with different configurations ($\mathbf{b}$, $\mathbf{t}$), $F_c(\alpha[001], [100]) = 0.55(0.33)$ eV/Å, $F_c(\alpha/2[111], [11\bar{2}]/\sqrt{6}) = 0.5(0.41)$ eV/Å and $F_c(\alpha/2[111], [1\bar{1}0]/\sqrt{2}) = 0.43(0.36)$ eV/Å [15]. Values out and in the brackets correspond to the choice of empirical potential [28] and [29,30], respectively. These values influence the total free energies of dislocation loops at varying temperatures,

Table 1

Fitting parameters for C_{11} , C' and C_{44} using the Varshni expression. Values in 2nd–4th columns were given by Adams [27] to predict experimental data in the temperature range of 3 – 500 K. Values in brackets are optimized ones in this work, which extends the prediction range to 1100 K.

Elastic constant	C_0 (GPa)	T_E (K)	S (GPa)	$f(T) = a_0 + a_1T + a_2T^2 + a_3T^3$			
				a_0	a_1	a_2	a_3
C_{11}	239.26	392.03	24.52 ($\times f(T)$)	1.93×10^{-1}	2.99×10^{-3}	-4.01×10^{-6}	2.20×10^{-9}
C'	51.75	371.00	9.06 ($\times f(T)$)	3.30×10^{-2}	4.34×10^{-3}	-6.42×10^{-6}	3.49×10^{-9}
C_{44}	120.72 (121.50)	162.90	3.50 (3.40)	–	–	–	–

suggesting that they accordingly affect the relative stability of dislocation loops.

Following the implication of core energies on loop stabilities, a set of F_c was fitted by Dudarev aiming to reach the consistency between experimental observations and the stability of dislocation loops reflected by their free energies [15]. In ref. [15], loops with a perimeter of 10 nm were used in the energy estimation, where loop sides are assumed to be parallel to dislocations with $\mathbf{b}$, $\mathbf{t} = \alpha[001]$, $[100]$; $\alpha/2[111]$, $[11\bar{2}]/\sqrt{6}$ and $\alpha/2[111]$, $[1\bar{1}0]/\sqrt{2}$. It should be pointed out that loop stability depends on not only the orientation of loop sides but also loop sizes. F_c obtained for loops with a specific perimeter of 10 nm might not be valid for other loops of different sizes.

To better evaluate F_c , we turn to the origin of loop formation for clues. Many simulations [19,22,31,32] have reported that prior to loop formation there exists a three-dimensional structure – C15 clusters – which is more stable than the two-dimensional dislocation loops containing the same number of SIAs. However, as C15 clusters grow they collapse into loops due to lower free energies of dislocation loops compared with C15 clusters. The collapse of C15 clusters is stochastic in the size of C15 clusters when the collapse occurs and in the type of loops formed after the collapse. Both $\alpha<001>$ and $\alpha/2<111>$ loops are possible products. It is reasonable to assume that the type of loops will depend on the free energies of loops of different configurations containing the same number of SIAs as the C15 cluster right before the collapse.

Zhang reported that the collapse of C15 clusters occurs at sizes of ~ 1.3 – 1.9 nm, equivalent to 57–147 SIAs [19]. Yao [16] found by TEM observation that in heavy-ion-irradiated pure iron, all the loops observed were $\alpha/2<111>$ type at 300 °C. The fraction of $\alpha<001>$ type loops increased with temperature. At 500 °C, all the loops were $\alpha<001>$ type and displayed square shapes [16,17]. These results indicate that at 300 °C and temperatures below, the free energy of $\alpha/2<111>$ loops containing 57–147 SIAs would be lower than $\alpha<001>$ loops containing the same number of SIAs. The trend reverses at 500 °C and higher temperatures.

Following the above observations and corresponding implications on loop stability, we fitted the zero-temperature F_c of dislocations with $\mathbf{b}$, $\mathbf{t} = \alpha[001]$, $<100>$; $\alpha/2[111]$, $[11\bar{2}]/\sqrt{6}$ and $\alpha/2[111]$, $[1\bar{1}0]/\sqrt{2}$ to satisfy the relation: $E_{tot}(\alpha/2<111>, 300^\circ\text{C}) < E_{tot}(\alpha<100>, 300^\circ\text{C})$ and $E_{tot}(\alpha/2<111>, 500^\circ\text{C}) > E_{tot}(\alpha<100>, 500^\circ\text{C})$ for loops containing 56–147 SIAs. With the fitted values (Table 2), the total free energy of loops with different Burgers vectors as a function of SIA contained and temperature can be computed. Fig. 3(a) shows that for a loop containing 100 SIAs, $\alpha/2[111]$ loops are definitely stable below 375 °C. At temperatures between 375 °C and 455 °C, $\alpha/2[111]$ loops with sides parallel to $\langle 1\bar{1}0 \rangle$ directions are thermodynamically unstable and tend to reduce their energy by either altering the orientation of their sides or by transforming into $\alpha[001]$ configurations. Above 455 °C, $\alpha[001]$ loops are unconditionally stable. Note that this result is valid only for loops containing 100 SIAs. The two critical temperature points (375 °C and 455 °C) would vary for loops of

Table 2

Values of core free energy at zero temperature extracted from atomistic calculations using different empirical potentials [28–30], and the fitted values in this work for three types of dislocations.

Dislocation type	Derived from atomistic calculations (eV/Å)		This work (eV/Å)
	Potential [28]	Potential [29,30]	
$\alpha[001]$, $[100]$	0.55	0.33	0.392
$\alpha/2[111]$, $[11\bar{2}]/\sqrt{6}$	0.5	0.41	0.446
$\alpha/2[111]$, $[1\bar{1}0]/\sqrt{2}$	0.43	0.36	0.429

different sizes. Fig. 3(b) shows critical temperatures for loops containing 56–147 SIAs. Three temperature regions with boundaries at 300 °C and 500 °C appear. In region 1 ($T < 300^\circ\text{C}$) loops formed via the collapse of unstable C15 clusters are the $\alpha/2[111]$ type. In region 2 ($300^\circ\text{C} < T < 500^\circ\text{C}$) loop type depends on the size of C15 clusters at the onset of the collapse. Clearly, with the increase of temperature $\alpha[001]$ loops are more likely to form. In region 3 ($T > 500^\circ\text{C}$) loops energetically prefer the $\alpha[001]$ configuration. This prediction perfectly agrees with the experimental findings by Yao [16]. We note also that the fitted values of zero-temperature F_c (Table 2) are well within the uncertainty levels predicted by the empirical potentials [28–30].

Fig. 4(a) shows the computed total free energy of loops with the configuration $\mathbf{b} = \alpha[001]$, $\mathbf{t} = \langle 100 \rangle$ as a function of the number of SIAs constituting the loop at various temperatures. Loop energy at small sizes generally agrees with the predictions by density functional theory (DFT) [10] and MD [19] calculations at zero temperature. The free energies tend to go down as temperature increases, and this trend aggravates at high temperatures (close to the α - γ phase temperature, 1185 K). Ultimately we aim to provide a simple analytical formula to estimate loop free energies, instead of applying the Eqs. (1–8). In previous works [10,33], free energies of loops at zero temperature have been parameterized by three terms: elastic energy $E_e(N) \sim \epsilon_0\sqrt{N}\ln N$, core energy $E_c(N) \sim \epsilon_1\sqrt{N}$ and a constant term ϵ_2 , where N is the number of SIAs. ϵ_0 , ϵ_1 and ϵ_2 are three adjustable parameters. Here we keep ϵ_0 and ϵ_1 only, and regard them as functions of temperature to account for temperature effects on loop free energy. For a loop with a given configuration containing N SIAs at temperature T , we assume that the free energy follows the formula

$$E_{tot}(N, T) = \epsilon_0(T)\sqrt{N}\ln N + \epsilon_1(T)\sqrt{N} \quad (10)$$

Values of $\epsilon_0(T)$ and $\epsilon_1(T)$ at different temperatures can be obtained by fitting free energies of loops to Eq. (10). $E_{tot}(N, T)$ of loops containing up to 20,000 SIAs ($R \sim 16$ nm for $\alpha[001]$ loops and ~ 17 nm for $\alpha/2[111]$ loops) at temperatures from 0 to 1100 K are calculated by Eq. (8) for the fits. To get a function explicitly containing T for $\epsilon_0(T)$ and $\epsilon_1(T)$, we propose a three-degree polynomial function

$$\epsilon_i(T) = b_0 + b_1T + b_2T^2 + b_3T^3, \quad (i = 0, 1), \quad (11)$$

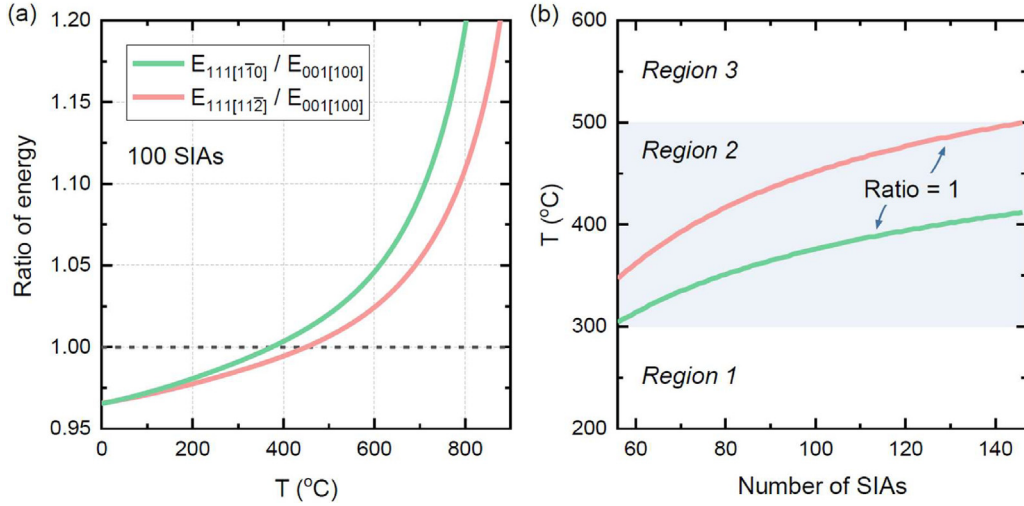


Fig. 3. (a) The ratio of the total free energies of loops containing 100 SIAs with $1/2[111]$ and $[001]$ Burgers vectors plotted as a function of temperature. (b) Critical temperature at which the ratio of the free energies for loops in (a) equals 1, plotted as a function of the number of SIAs constituting the loop. Above the curve, the corresponding ratio is larger than 1 and vice versa for the ratios below the curve.

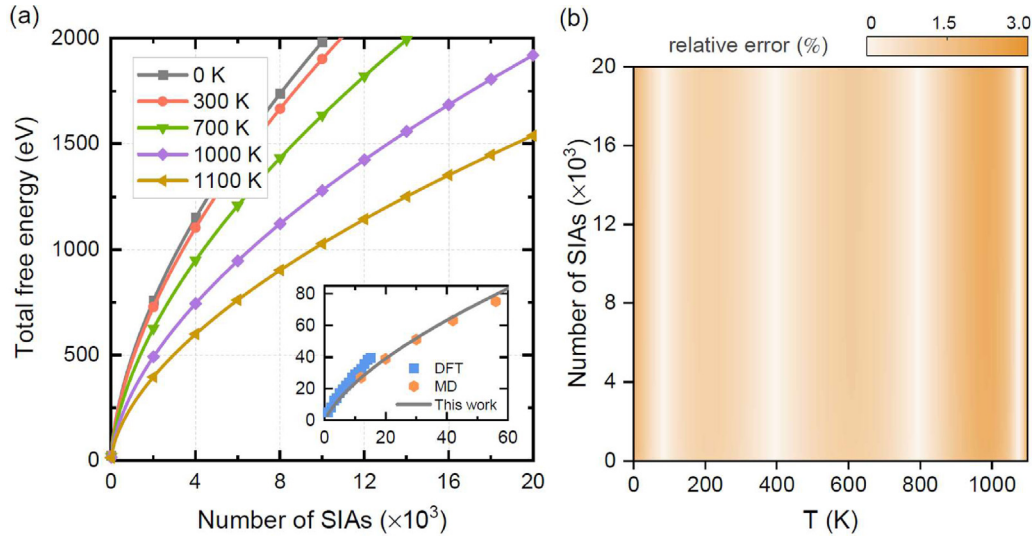


Fig. 4. (a) Total free energy of dislocation loops with $[001]<100>$ configuration at different temperatures plotted as a function of the number of SIAs constituting the loop. The inset shows the comparison of energies calculated by DFT [10], MD [19] and anisotropic elasticity for small loops at 0 K. (b) Relative error between the free energies calculated based on dislocation elasticity and by analytical formulas.

Table 3

Best-fit coefficients b_j ($j = 0-3$) for $\epsilon_0(T)$ and $\epsilon_1(T)$ according to Eq. (11). Prismatic dislocation loops with Burgers vectors $[001]$ (loop sides parallel to $\langle 100 \rangle$ directions), and $1/2[111]$ (loop sides parallel to $\langle 11\bar{2} \rangle$ or $\langle 1\bar{1}0 \rangle$ directions) are evaluated.

Coefficients	$[001]$		$1/2[111]$			
	$\langle 100 \rangle$		$\langle 11\bar{2} \rangle$		$\langle 1\bar{1}0 \rangle$	
	$\epsilon_0(T)$	$\epsilon_1(T)$	$\epsilon_0(T)$	$\epsilon_1(T)$	$\epsilon_0(T)$	$\epsilon_1(T)$
b_0	1.850	3.180	1.580	3.963	1.601	3.788
b_1	-5.716×10^{-4}	-4.356×10^{-4}	-4.482×10^{-4}	-6.033×10^{-4}	-2.930×10^{-4}	-3.973×10^{-4}
b_2	7.044×10^{-7}	-3.227×10^{-8}	5.519×10^{-7}	2.639×10^{-7}	1.218×10^{-7}	-2.372×10^{-7}
b_3	-8.752×10^{-10}	-4.002×10^{-10}	-6.338×10^{-10}	-5.344×10^{-10}	-2.740×10^{-10}	-1.247×10^{-10}

The coefficients b_j ($j = 0-3$) generated from best fits are listed in Table 3. Fig. 4(b) shows the relative error between free energies evaluated by Eqs. (8) and (10) for $\alpha[001]$ loops with sides parallel to $\langle 100 \rangle$ directions. The maximum error is 2.21% around 988 K. For $\alpha/2[111]$ loops, maximum relative errors are 1.05% and 1.74% for loop sides parallel to $\langle 11\bar{2} \rangle$ and $\langle 1\bar{1}0 \rangle$ directions, respectively.

In summary, this work employs the anisotropic elasticity of dislocations in bcc iron to assess the free formation energies of pris-

matic dislocation loops as a function of temperature. Based on available experimental observations, a new scaling law explicitly containing temperature is established for the free energies of loops up to 30 nm in diameter and 1100 K in temperature. The analytical formula along with the fitted parameters may have a wide application in multiscale simulations such as kinetic Monte Carlo simulations, cluster dynamics studies, or mean-field approximations for loop dynamics in iron under irradiation or annealing.

Credit author statement

J.G. wrote the code, performed the calculation and data analysis, and drafted the manuscript. E.G. supervised the entire work. E.G. and J.A. commented on the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge helpful discussions with Dr. S.L. Dudarev. This work is supported by the [Helmholtz Association](#) of German Research Centers (HGF) and has been carried out within the framework of Nuclear Fusion Programme at Karlsruhe Institute of Technology (KIT), Germany.

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